



Acute Stroke

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Introduction

Stroke, recognized clinically as the sudden onset of a focal disturbance of central nervous system function, is caused by cerebral infarction (ischemic stroke, responsible for about 85% of all strokes) or intracerebral hemorrhage.^[1] Subarachnoid hemorrhage, the most common type of hemorrhage, more often causes sudden onset of severe headache with or without impaired consciousness and/or focal findings on neurological examination.

A transient ischemic attack (TIA) is a temporary period of symptoms similar to those caused by an ischemic stroke, and not resulting in lasting deficits. TIAs are clinically very important, as they can indicate the risk of future stroke in up to one-third of people. TIAs should be viewed as an opportunity to prevent future strokes and must be treated as an emergency.

The Heart and Stroke Foundation of Canada promotes awareness of the [warning signs of possible stroke](#) (see [Table 1](#)) to encourage prompt response when needed.

Table 1: Warning Signs of Stroke

F.A.S.T.	
Face—Is it drooping?	Speech—Is it slurred or jumbled?
Arms—Can you raise both?	Time—Time to call 911 right away

Goals of Therapy

- Minimize brain damage
- Prevent complications resulting from stroke-related deficits, e.g., pneumonia, venous thromboembolism
- Reduce risk of stroke recurrence
- Restore function and minimize long-term disability for the individual

Investigations

The evaluation of patients with suspected stroke is highly time-dependent. Specific stroke treatments (recanalization with thrombolytic drugs and/or endovascular thrombectomy) can be provided only within the first hours of symptom onset in most patients with ischemic strokes (see [Figure 1](#), [Figure 2](#)).

Use the clinical history, physical examination, imaging studies and other ancillary investigations to help confirm the diagnosis and exclude stroke mimics such as subdural hematoma, Todd paralysis (post-seizure), brain abscess, herpes simplex encephalitis, hypoglycemia, brain tumor (primary or secondary), multiple sclerosis, migraine and conversion disorder.^[2]

- History:
 - time last seen well, time of onset, if symptoms noted upon waking from sleep
 - symptoms at onset, neurologic worsening or improvement since onset
 - medications (particularly time of last dose of any anticoagulant or antiplatelet drugs as well as drugs that might hint at other relevant conditions—anticonvulsants, diabetes medications, chemotherapy drugs, steroids)
 - antecedent trauma, illness or previous neurovascular events
 - vascular comorbidity (angina, MI, heart failure, peripheral and renal vascular disease) or atrial fibrillation
 - vascular disease risk factors (hypertension, smoking, diabetes mellitus, dyslipidemia, excessive alcohol intake, high body mass index, low exercise, migraine with aura, obstructive sleep apnea, family history of vascular disease or hemostatic disorders)
 - pre-stroke cognitive and functional status
 - place of residence, occupation and social supports
- Physical examination:
 - rapid evaluation of airway, breathing and circulation; vitals including BP should be obtained
 - localize lesion and determine stroke severity by using the National Institutes of Health Stroke Scale (NIHSS)
 - assess comorbid conditions and determine possible cause of stroke
- Emergency laboratory and radiological tests:

- CBC, INR, PTT, albumin, electrolytes, glucose (to rule out hypoglycemia as a stroke mimic), HbA_{1c}, urea, creatinine, troponin, liver function tests, lipid profile and TSH
- immediate neurovascular imaging to confirm the clinical diagnosis and plan treatment; urgent noncontrast CT head and CT angiography (CTA) of head and neck is the preferred imaging modality in the hyperacute phase of stroke:
 - noncontrast CT head can demonstrate changes associated with ischemic strokes, hemorrhagic strokes or stroke mimics, such as tumors
 - CTA head can show intracranial occlusions or stenosis; multiphase CTA head can provide information about the state of collateral circulation in the brain at time of the stroke and assist in acute management
 - CTA neck may demonstrate occlusion or stenosis of carotid, vertebral or basilar arteries, or may indicate presence of a vascular dissection or other abnormalities such as fibromuscular dysplasia; patients with carotid stenosis may be amenable to revascularization therapy if they have had a TIA or nondisabling stroke
 - CT perfusion is sometimes used; it can improve the diagnostic yield of small ischemic strokes, rule out mimics and help with treatment decisions
 - MRI brain is occasionally used for extended-window thrombolysis decisions; however, emergency access to MRI in Canada is limited
 - carotid dopplers remain a useful screening tool for carotid stenosis if CTA not available
- ECG (to look for atrial fibrillation, MI and left ventricular hypertrophy)
- Other investigations if indicated:
 - chest x-ray (to look for heart disease, lung cancer)
 - prolonged ECG monitoring (minimum 24 h Holter monitor, but 2 wk recommended)^[3] and echocardiography (transthoracic [with bubble study in patients <60] ± transesophageal) to search for atrial fibrillation or other cardiac source of emboli in patients with embolic strokes
 - brain MRI (+ diffusion weighted imaging) if cause of stroke is uncertain, particularly in young patients
 - antiphospholipid antibodies, factor V Leiden and prothrombin gene mutation if arterial hypercoagulable state suspected; protein C, protein S, antithrombin III could be added if venous embolism is suspected in the case of a younger person with a confirmed patent foramen ovale on echocardiography
 - inflammatory markers (ESR, CRP, antinuclear antibody [ANA]) as a screen for vasculitis
 - blood cultures and urgent echocardiogram if infectious endocarditis suspected
 - MRA with vessel wall imaging +/- catheter cerebral angiography if vasculitis or vascular anomaly suspected
 - lumbar puncture if suspected vasculitis, syphilis or other CNS infection
 - if CT is negative for blood but subarachnoid hemorrhage is still suspected, lumbar puncture is indicated
 - malignancy workup (CT chest, abdomen, pelvis)
 - pregnancy test
 - syphilis and HIV serology

Therapeutic Choices

Nonpharmacologic Choices

Endovascular Thrombectomy for Acute Ischemic Stroke

Patients who have had major anterior circulation ischemic strokes with specific features (see [Table 2](#)) stand to benefit from endovascular thrombectomy (EVT).^[4] EVT is a nonsurgical, angiographic procedure that uses microcatheters to remove blood clots and re-establish arterial blood flow in the brain. [Figure 1](#) and [Table 2](#) suggest a strategy to balance the potential benefit of EVT against persisting uncertainties and resource considerations. The time window for intervention is evolving, as we understand more about tissue-window in stroke care. Studies have shown that select patients identified using specialized imaging modalities (including automated calculation of infarct core as well as viable brain tissue represented by the ischemic penumbra) can benefit from EVT up to 24 hours after stroke onset.^{[5][6][7]} Select patients with stroke caused by basilar artery occlusion should also be considered for EVT.^{[8][9][10][11]}

EVT is best achieved by a clinician experienced in the procedure using a stent retriever and/or contact aspiration device.^[12] EVT may be performed in patients who have or have not been treated with thrombolytics (see [Table 3](#)).

Table 2: Indications for Endovascular Thrombectomy^{[2][13]}

All must apply:

- Functionally disabling stroke with deficits associated with proximal cerebral artery occlusion: hemisensorimotor signs, aphasia or neglect, with or without homonymous hemianopia
- Small-to-moderate ischemic core defined by an Alberta Stroke Program Early CT Score (ASPECTS) of 6 or more on noncontrast CT or CT perfusion (cerebral blood volume map)
- Intracranial proximal artery occlusion in anterior circulation (distal ICA or proximal middle cerebral artery [M1] or proximal anterior cerebral artery [A1] on CTA); vertebrobasilar occlusions may also be considered, although evidence base is much less certain in these cases
- Moderate-to-good pial collateral filling on multiphase CTA or evidence of CT perfusion mismatch
- Aortic arch that is navigable
- Treatment can be started within 6 h of symptom recognition *and* within 12 h of when patient last known to be well

Special circumstances:

- For basilar artery occlusions, decision to treat with EVT should be based on the potential risks and benefits, but studies showed benefit^{[8][9][10] [11]}
- Highly selected patients who arrive later than 6 h after stroke onset may benefit from EVT if there is imaging and clinical evidence of small core and a large area at risk^{[5][6]}
- Although benefits are uncertain, some patients with anterior circulation large vessel occlusions with ASPECTS <6, NIHSS score <6 or mRS >1 may benefit from EVT^[13]

Abbreviations: ACA = anterior cerebral artery; CT = computed tomography; CTA = CT angiography; EVT = endovascular thrombectomy; ICA = internal carotid artery; MCA = middle cerebral artery; mRS = modified Rankin Scale; NIHSS = National Institutes of Health Stroke Scale

Neurosurgical Intervention

- Ischemic stroke
 - Decompressive hemicraniectomy may be performed on young patients with massive hemispheric infarction associated with significant edema and mass effect (“malignant MCA syndrome”).^[2] Evidence from a pooled analysis of 3 small randomized trials found an increase in survival for patients under 60 years of age when treated with craniectomy; however, survivors incurred significant disability resulting in an inability to live independently. Those involved in decision-making should be aware of this significant trade-off.^[14]
 - Consensus opinion is that patients with large cerebellar infarcts may benefit from surgical decompression.^[2]
- Intracerebral hemorrhage (see [Figure 3](#))
 - Patients with cerebellar hemorrhage should have an urgent neurosurgical consultation for consideration of posterior fossa craniotomy and evacuation of the hemorrhage.^[2] The role of neurosurgical intervention for supratentorial intracerebral hemorrhage is unclear and the subject of ongoing research.^{[15][16]}
- Subarachnoid hemorrhage (see [Figure 3](#))
 - Refer patients with suspected subarachnoid hemorrhage immediately to a neurosurgical centre.
 - For aneurysmal subarachnoid hemorrhage, ablation of the aneurysm reduces the risk of re-bleeding. Endovascular coiling is superior to neurosurgical clipping for a ruptured intracranial aneurysm that is anatomically suitable for either procedure.^[17]

General Supportive Care

- Outcomes are optimized by care on a stroke unit provided by a coordinated interdisciplinary team (see [Figure 4](#)).^[18]
- Supplemental oxygen is not required for patients with normal oxygen saturation levels.^[2]

- Elevated body temperature is associated with poor outcome after stroke.^[19] Symptomatic treatment of pyrexia (and investigation of its cause) is recommended.^[2]
- Begin rehabilitation as soon as patient is medically stable.^[20]
- Family and community supports are important for social reintegration.^[20]

Managing Swallowing and Nutrition^[20]

- Give nothing by mouth if any of the following are present: reduced level of consciousness, suspected brainstem stroke, severe dysarthria, wet voice, weak cough, impaired palatal sensation, inability to sit, suspected aspiration.
- Bedside dysphagia screen should be performed soon after a stroke; this is best performed by an experienced dysphagia team.
- A videofluoroscopic examination (modified barium swallow) may be required to exclude significant aspiration when the results of the bedside examination are ambiguous.
- Monitor recovery using serial bedside swallowing assessments +/- additional fluoroscopic assessments as needed.
- A texture-modified diet is appropriate for dysphagic patients at lower risk of aspiration. Additional IV fluids are often necessary for these patients.
- Tube feeding (usually nasogastric tube) may be required if significant aspiration is demonstrated or suspected and this is within the goals of care for the patient (some patients may choose to “eat at risk of aspiration”).
- If a nasogastric tube cannot be tolerated or cannot be placed, consider short-term partial parenteral nutrition or total parenteral nutrition.
- If swallowing does not improve, percutaneous endoscopic gastrostomy (PEG) should be discussed with dysphagia experts. Early initiation of PEG feeding in dysphagic stroke patients is not recommended.^[21]

Preventing Venous Thromboembolism^[2]

- Early mobilization is recommended (as soon as possible after admission to hospital).
- Maintain adequate hydration.
- Intermittent pneumatic compression stockings should be applied as soon as possible within the first 24 hours to reduce the risk of deep vein thrombosis and increase the likelihood of survival after stroke; because they do not cause bleeding, they can be used in patients who have had an intracerebral hemorrhage.^[22] The use of graduated compression stockings is not recommended because of a lack of efficacy.

Pharmacologic Choices

Ischemic Stroke

Thrombolytics

Administer IV **alteplase** (rt-PA) as soon as possible following stroke onset in patients who meet eligibility criteria (see [Figure 1](#), [Table 3](#), [Table 4](#) and [Table 5](#)).^[2] A meta-analysis of individual patient data from 9 randomized trials involving 6756 participants selected on the basis of their clinical presentation and the findings on a noncontrast CT brain scan showed that thrombolysis within 3 hours significantly increased the chance (from 231–329 per 1000 patients treated) of being alive and independent 3–6 months after ischemic stroke.^[23] Thrombolysis between 3 and 4.5 hours provided a smaller yet still significant benefit (chance of independent survival increased from 301 to 353 per 1000 patients treated). This overall benefit was found despite an increase in early intracranial hemorrhage (5.4% increase compared to the control in the NINDS trial).^[24] Select patients who cannot have EVT may benefit from alteplase within 9 hours of stroke onset. The WAKE UP and EXTEND trials showed alteplase was beneficial in carefully selected patients even when administered after >4.5 hours.^{[25][26]}

Tenecteplase (TNK) may be considered as an alternative thrombolytic agent to alteplase in stroke treatment. It has greater fibrin specificity, longer half-life, and is administered as a single bolus rather than a one-hour infusion. TNK at 0.25 mg/kg (as a single bolus, maximum of 25 mg) has been shown to be noninferior to alteplase in efficacy and safety; in the ACT trial, similar proportion of patients achieved excellent outcome (36.9% in the TNK group and 34.8% in the alteplase group) without an increase in symptomatic intracranial hemorrhage (3.4% in the TNK group and 3.2% in the alteplase group).^{[27][28]} This benefit has been replicated in other studies.^{[29][30]} The use of TNK for treatment of acute ischemic strokes was approved by Health Canada at the end of 2025. Most Canadian centres have switched from using rt-PA to TNK for treatment of acute ischemic strokes.

Table 3: Thrombolysis in Acute Ischemic Stroke: Canadian Treatment Criteria^[2]

Treatment criteria	- Ischemic stroke causing measurable neurologic deficit in a patient ≥18 y - Stroke onset ≤4.5 h before drug administration
Exclusion criteria	Absolute - Active hemorrhage or any condition that could increase the risk of major bleeding after thrombolytic administration - Any acute hemorrhage on intracranial imaging Relative (requiring clinical judgment) Historical - history of intracranial hemorrhage - stroke or serious head or spinal trauma within the preceding 3 months - major surgery in preceding 14 days - arterial puncture at a noncompressible site in the preceding 7 days

- any other condition that could increase the risk of hemorrhage after thrombolytic administration
- **Clinical**
 - symptoms suggestive of subarachnoid hemorrhage
 - stroke symptoms due to another nonischemic acute neurologic condition such as seizure with postictal Todd paralysis, or severe hypo- or hyperglycemia
 - hypertension refractory to therapy such that target BP <180/105 mm Hg cannot be reached
 - patient taking a DOAC
- **Laboratory**
 - blood glucose concentration <2.7 or >22.2 mmol/L
 - elevated activated partial thromboplastin time
 - INR >1.7
 - platelet count <100 × 10⁹/L
- **CT or MRI Findings**
 - scan showing signs of extensive infarction (due to treatment futility and/or higher rates of hemorrhagic transformation)

Abbreviations: BP = blood pressure; CT = computed tomography; DOAC = direct-acting oral anticoagulant; INR = international normalized ratio; MRI = magnetic resonance imaging

Complications of Intravenous Thrombolysis

- **Intracerebral hemorrhage (ICH).** Clinically suspected if there is a sudden deterioration in the patient's neurological status. Incidence of symptomatic ICH is 3–6%.^{[24][28]} Monitor vital signs. Stop infusion and obtain a STAT CT head to rule out ICH. If an ICH is present, it may be reasonable to discuss the case with a neurosurgeon, but there is no randomized controlled trial evidence to support surgical intervention. Supportive care according to the patient's goals of care is recommended.
- **Allergic reaction (hemilingular angioedema).**^[31] A small proportion (<5%) of patients who have been taking angiotensin-converting enzyme inhibitors (ACEi) will experience hemilingular angioedema when treated with IV thrombolysis; this can be life-threatening. Clinical observation for signs of airway obstruction should be a standard of care following IV thrombolysis. Signs of angioedema (tongue swelling, stridor) should initially be managed by clinicians skilled in airway management. Pharmacological management includes IV steroids and antihistamines.

Table 4: Thrombolysis in Acute Ischemic Stroke: Monitoring^[13]

BP and Neurologic Signs

- Baseline, then Q15 min × 2 h after starting alteplase, then Q30 min × 6 h, then Q1H until 24 h after starting thrombolytic
- Call MD if systolic BP is >180 mm Hg or if diastolic BP is >110 mm Hg on 2 or more occasions taken 5–10 min apart
- Stop the infusion, obtain emergency CT scan and notify MD if there is neurologic deterioration, severe headache, or new onset of nausea or vomiting

Bleeding

- Visual assessment for signs of bleeding Q30 min during infusion and Q1H x 6 h, then Q4H until 24 h after starting thrombolytic

Blood Glucose

- Call MD if glucose >12 mmol/L

Lines and Tubes

- Delay placement of nasogastric tubes, indwelling catheters or intra-arterial pressure catheters

Medications

- No anticoagulant or antiplatelet drug for 24 h and until repeat CT scan rules out hemorrhagic transformation
- Acetaminophen 650 mg PO or PR Q4H if body temperature is ≥38°C or for analgesia
- O₂ via nasal prongs or face mask to keep O₂ saturation >90%
- After completion of thrombolytic infusion, continue IV normal saline (with or without KCl)

Investigations

- CT brain scan after 24 h

Abbreviations: BP = blood pressure; CT = computed tomography; IV = intravenous; KCl = potassium chloride; MD = medical doctor

Antiplatelet Therapy^[2]

- **ASA** is administered as follows:
 - If intracranial hemorrhage is excluded by CT scan but thrombolysis is not indicated, give ASA 160 mg immediately. Follow with ASA 80–325 mg daily indefinitely.^{[32][33]}

- When IV thrombolysis has been administered, wait until intracranial hemorrhage is excluded by CT scan 24 hours after treatment,^[34] then give ASA 160 mg once followed by ASA 80–325 mg daily indefinitely.
- Following EVT, the timing of ASA administration is still debated, but many clinicians will wait until after follow-up imaging has excluded an ICH before starting ASA 80–325 mg daily indefinitely.
- Administer ASA as a suppository or via nasogastric tube to dysphagic patients. No evidence supports the use of ASA doses greater than 325 mg/day for secondary stroke prevention. The GI side effects of ASA are dose-related.
- For patients who were taking ASA prior to their stroke, alternate antiplatelet agents, such as **clopidogrel** 75 mg once daily, or **ticagrelor** 90 mg twice daily can be considered. These regimens have not been tested following acute stroke, but have been studied in long-term secondary prevention trials and have been found to be superior to ASA and equivalent to each other in ASA-naïve patients^[3] (see [Prevention of Ischemic Stroke](#)).
- For those with TIAs or minor strokes who are not at high risk of bleeding (e.g., no history of significant GI or genitourinary bleeding), the use of dual antiplatelet therapy for the first 3–4 weeks is recommended.
 - The use of combined ASA and clopidogrel in the first 3–4 weeks after a nondisabling stroke (or a TIA) reduces the risk of major stroke recurrence at 3 months without increasing the likelihood of serious bleeding.^{[35][36]}
 - ASA and ticagrelor combination therapy for 3–4 weeks may be considered as an alternative.^[37] Additionally, there is strong evidence in support of ticagrelor and ASA use in those with clopidogrel resistance, characterized by CYP2C19 loss-of-function allele.^[38] The prevalence of poor clopidogrel metabolism is 2–15% depending on patients' ethnic background.^[39] Pharmacogenomic testing is not widely commercially available in Canada at this time.
- Long-term dual antiplatelet therapy may be used in select patients (for example, in combination with statins in patients with intracranial atherosclerosis),^[40] but dual-antiplatelet therapy is generally not recommended for long-term secondary stroke prevention due to risk of bleeding without significant benefit.^{[41][42][43][44]}

Anticoagulant Therapy

- Systemic anticoagulation initiated immediately after acute ischemic stroke is not routinely recommended.^[45] There may be special circumstances when this is indicated (e.g., venous infarcts caused by cerebral venous sinus thrombosis or carotid artery free floating thrombus). For patients with atrial fibrillation (AF), immediate anticoagulation results in reduction in early recurrent ischemic stroke, but this benefit is offset by an increase in major intracranial and extracranial bleeding.^[45] For patients with AF and ischemic stroke, most physicians use **ASA** until an oral anticoagulant is started.^[3]
- For patients with AF or other cardioembolic causes of stroke, consider long-term anticoagulation using **apixaban**, **dabigatran**, **edoxaban** or **rivaroxaban** in preference to **warfarin** (at a dose to maintain the INR in the range of 2–3).^[3] DOACs are preferred because of reduced risk of bleeding (including intracranial hemorrhage), lack of INR monitoring to ensure the drug is within therapeutic range and fewer drug interactions. For patients who cannot be anticoagulated, use enteric-coated ASA 80–325 mg daily.^[3]

- The choice of oral anticoagulant should be based on patient factors including age, renal function, additional health factors, drug interactions, likelihood of adherence and patient preferences.
- The best time to initiate anticoagulant therapy is unclear. The “1-3-6-12 day rule” is a known consensus opinion to guide initiation of anticoagulation for nonvalvular AF. Patients with TIA would re-institute anticoagulation after 1 day. Patients with mild, moderate or severe stroke, most often based on the NIHSS score, would wait 3, 6 or 12 days, respectively.^[46] Recent studies have shown it is safe and beneficial to initiate anticoagulation earlier, typically within 5 days.^{[47][48][49]} In practice, it is common to repeat brain imaging after 1-14 days to exclude asymptomatic intracranial hemorrhage before starting anticoagulation.

For information on postacute antithrombotic treatment, carotid endarterectomy and risk factor modification, see [Prevention of Ischemic Stroke](#).

Preventing Venous Thromboembolism^[2]

There is evidence for both intermittent pneumatic stockings (IPS) or chemoprophylaxis for DVT/PE prevention in acute stroke patients, but there is no data to support the superiority of either approach. In the absence of contraindications, use of **low molecular weight heparin (LMWH)** or **unfractionated heparin (UH)** in prophylactic doses (see [Venous Thromboembolism](#)) can be considered in patients with acute ischemic stroke who cannot move one or both lower limbs or mobilize independently.^[50] Prophylaxis should be stopped once an anticoagulant is prescribed.

Blood Glucose Management

Hypoglycemia should be corrected immediately. Hyperglycemia is associated with poor outcome after stroke. It is a risk factor for intracerebral hemorrhage after thrombolytic therapy and should be treated.^[2] Using a SC insulin regimen is recommended since aggressive management of blood glucose using IV insulin did not improve outcomes and was associated with more hypoglycemia.^[51]

Blood Pressure-Lowering Treatment

- In patients eligible for alteplase, opinion-based guidelines recommend treating elevated BP to below 185/110 mm Hg before alteplase is given.^[2] The optimum BP target for ischemic stroke patients treated with alteplase has not been established. ENCHANTED, an international multicentre trial, enrolled patients with mild to moderately severe ischemic stroke treated with alteplase and randomly allocated them to intensive BP reduction (systolic BP 130–140 mm Hg) versus guideline (systolic target <180 mm Hg).^[52] The trial showed intensive BP lowering did not improve functional outcome despite a lower rate of intracranial hemorrhage.
- Similarly, optimum BP after EVT is an area of ongoing research.

Hemorrhagic Stroke

Intracerebral Hemorrhage^[53]

The mainstay of intracerebral hemorrhage treatments are BP control and anticoagulant reversal, but the effect size of these interventions is small.^[53]

Treat patients with acute intracerebral hemorrhage and an established coagulopathy or a history of warfarin use with **prothrombin complex concentrate**, **vitamin K** or fresh-frozen **plasma** to reverse the coagulopathy. **Idarucizumab** reverses the anticoagulant effect of dabigatran.^[54] **Andexanet alfa** is a reversal agent for factor Xa inhibitors (apixaban, edoxaban and rivaroxaban) that is not widely available in Canada.^[55] Andexanet alfa has been studied in acute intracerebral hemorrhage patients who are on factor Xa inhibitors. It showed hemostatic efficacy in 14% more patients compared to standard therapy, but there was an increase in thrombotic events.^[56] If there is a persisting strong indication for anticoagulation (e.g., mechanical heart valve), decide when to restart anticoagulant therapy on a case-by-case basis. Administration of **recombinant factor VIIa** prevents hematoma growth, but is not recommended for routine use because it increases the risk of arterial thromboembolic phenomena and does not reduce death or disability.^[57] Its use in hyperacute ICH (<2 h from onset) is an area of ongoing research.

Intensive BP lowering with IV agents to <140 mm Hg systolic (compared with <180 mm Hg target) in patients with acute intracerebral hemorrhage is safe but is not associated with clearly improved functional outcome at 3 months post-stroke.^{[58][59]} INTERACT2 showed no effect on mortality or severe disability, but improved functional outcome for intensive BP lowering using the ordinal shift analysis. Ten years later, INTERACT3 showed improved functional outcomes for intensive BP lowering as part of a care bundle that included treatment of hyperglycemia, pyrexia and warfarin reversal, mainly in low- and middle-income countries.^[60] In practice, lowering systolic BP to less than 140–160 mm Hg may be reasonable, particularly in patients who present within the first 6 hours with severe hypertension, have imaging markers of hematoma expansion (e.g., spot sign) and have normal renal function.

After the acute phase, manage patients with intracerebral hemorrhage similarly to those with ischemic stroke, except avoid anticoagulants.

In patients with hemorrhagic stroke, the use of LMWH or UH for venous thromboembolism prophylaxis is controversial. Intermittent pneumatic stockings should be applied within 24 h, but chemoprophylaxis can be considered 48 h after hematoma stabilization.

Resumption of long-term antithrombotic therapy in this population is an area of ongoing research. RESTART showed treatment with a single antiplatelet agent in ICH patients who have an indication (e.g., coronary artery disease) is safe and does not increase the risk of recurrent intracerebral hemorrhage.^[61] The safety of anticoagulation in patients with ICH and atrial fibrillation was indeterminate in SOSTART and APACHE-AF.^{[62][63]} This question is being examined further in several randomized trials.

Intracerebral hemorrhage is not an indication for statin therapy. The indication for statin use should be reviewed if a patient is using a statin; in a trial that included patients with hemorrhagic as well as ischemic stroke, statins use was correlated with a higher risk of intracerebral hemorrhage.^[64]

Subarachnoid Hemorrhage

Treatment of subarachnoid hemorrhage is primarily nonpharmacologic (ablation of the bleeding source). Refer patients with suspected subarachnoid hemorrhage to a neurosurgical centre immediately.

Nimodipine 60 mg PO Q4H for 3 weeks reduces the risk of secondary vasospasm and cerebral infarction. Consider a dose of 30 mg PO Q2H in patients who are very sensitive to the BP lowering effects (see [Table 6](#)).^[65]

Choices During Pregnancy and Breastfeeding

Acute Stroke and Pregnancy

Acute stroke management during pregnancy is the topic of a best practice consensus statement.^[66] A Canadian study reported a stroke frequency of 13.4 per 100 000 deliveries, which is at least double that seen in the general population of young adults.^[67] Stroke during pregnancy is more likely to be hemorrhagic (about 59% of cases) than ischemic and most often occurs close to the time of delivery and in the postpartum period (about 52% of cases).^[68]

Risk factors for stroke in pregnancy are older age, preeclampsia and eclampsia, HELLP (hemolysis, elevated liver enzymes and low platelets) syndrome, congenital heart disease, connective tissue disorders, sepsis, severe postpartum hemorrhage, and thrombophilic disorders.^[68]

Other neurovascular ("stroke overlap") syndromes, such as reversible cerebral vasoconstriction syndrome and posterior reversible encephalopathy syndrome, may also be associated with severe preeclampsia.

Management

An interdisciplinary team approach is preferred to address care and management decisions and should involve those with stroke, obstetrical and maternal-fetal medicine expertise.

Initial emergency investigations and treatment of stroke during pregnancy or within the 6 weeks after delivery are similar to the approach taken in the nonpregnant patient.

There is insufficient evidence to support the safety of antiplatelet agents other than **ASA** in pregnancy. There are insufficient data on the safety of direct oral anticoagulants in pregnancy. Warfarin is potentially teratogenic and should be avoided, especially between 6 and 12 weeks' gestational age. When anticoagulation is considered, LMWH is preferred throughout pregnancy.

For intracerebral hemorrhage, the priorities are BP control and the identification and management of coagulopathies.

Imaging Considerations

A typical noncontrast CT scan of the pregnant patient's head exposes the fetus to a negligible amount of radiation. There is a lack of evidence on, but no known harm from, CT contrast. MRI is not associated with fetal risk, but gadolinium enhancement should be avoided.^[69]

Thrombolytic Therapy

Pregnant patients were excluded from the trials of alteplase therapy for acute ischemic stroke; however, case reports suggest that the benefits of therapy outweigh the risks. Guidelines advise that **alteplase** is a reasonable treatment option.^[66]

Patient health considerations are that alteplase may increase the risk of postpartum hemorrhage, bleeding after epidural or spinal anesthesia, and bleeding following cesarean delivery.

Alteplase and tenecteplase do not cross the placenta.^{[70][71]} Therefore, they are not expected or known to pose any direct risk to the fetus, but studies of the use of TNK in pregnancy are limited.

Endovascular Thrombectomy

A pregnant patient with a large vessel occlusion should be offered EVT. Depending on the clinical scenario, it may be reasonable to forgo thrombolysis.

Acute Stroke and Breastfeeding

Initial emergency investigations and treatment of stroke within the 6 weeks after delivery are similar to the approach taken in the nonpregnant patient. Continuation of breastfeeding after exposure to CT contrast does not pose any known risk to the infant.

Thrombolysis may increase the risk of postpartum hemorrhage, bleeding after epidural or spinal anesthesia, and bleeding following cesarean delivery. Any thrombolytic drug is likely destroyed in the infant's GI tract. There is probably no hazard to the infant of a breastfeeding patient treated with thrombolysis, though no information is available on the clinical use during breastfeeding.^[70]

Low-dose **ASA** can be considered during breastfeeding since there is evidence that it is not transferred into breast milk, and salicylate blood levels are low in patients taking daily low-dose ASA.

If anticoagulation is required, **LMWH** and **warfarin** are both considered safe options during breastfeeding. The safety of DOACs in breastfeeding have not been established.

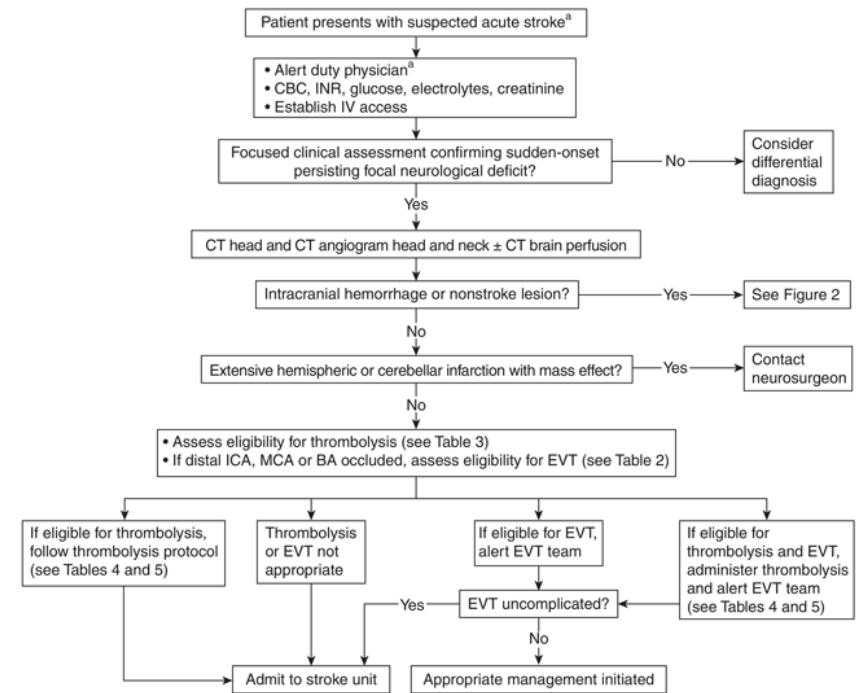
A discussion of general principles on the use of medications in these special populations can be found in [Drugs Use during Pregnancy](#), and [Drug Use during Breastfeeding](#). Other specialized reference sources are also provided in these appendices.

Therapeutic Tips

- The effectiveness of IV thrombolytic therapy and EVT are exquisitely time-dependent; avoid delays. Immediate contact with the patient, rapid triage and (most important) staying with the patient continuously during the clinical assessment, CT scan, blood tests and consent procedures are vital in ensuring that the appropriate steps are being taken as rapidly as possible prior to thrombolytic administration.
- Determining the time of stroke onset is critical in deciding to treat with IV thrombolytics and EVT, but checking the clock is not a natural reaction in the setting of an acute stroke. Encourage patients and families to think of “time anchors” (e.g., what was on the radio or TV at the time, or at what point in the patient’s daily routine the symptoms first began).
- Patients with acute stroke are often unable to communicate. When possible, the next of kin should travel with the patient to provide collateral history and consent for treatment before the time window for intervention closes.
- If the patient is referred to a tertiary-care hospital, send the registration page and results of the STAT blood work (CBC, INR) drawn at the community hospital to the referral centre.
- Waiting for INR and PTT may not be necessary if the patient has no medical condition that affects blood clotting and is not on anticoagulant therapy.
- Signs of infarction on a CT scan done within 4.5 hours of stroke onset are usually subtle. If the CT scan of a patient being considered for acute treatment shows a very definite infarct in a location that explains the presenting clinical symptoms and signs, recheck the time of onset. If the time of onset is accurate, then the patient likely has poor collateral circulation.

Algorithms

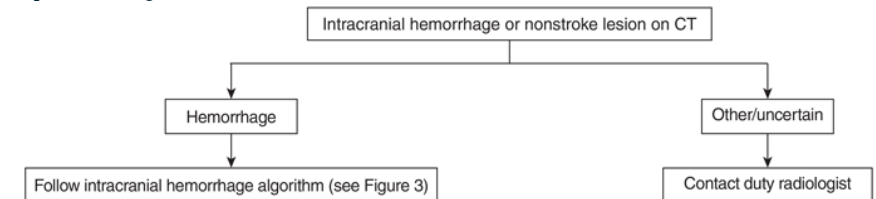
Figure 1: Emergency Department Management of Patients with Suspected Stroke



^[a] Facilitated by pre-hospital systems and protocols.^{[2][3]}

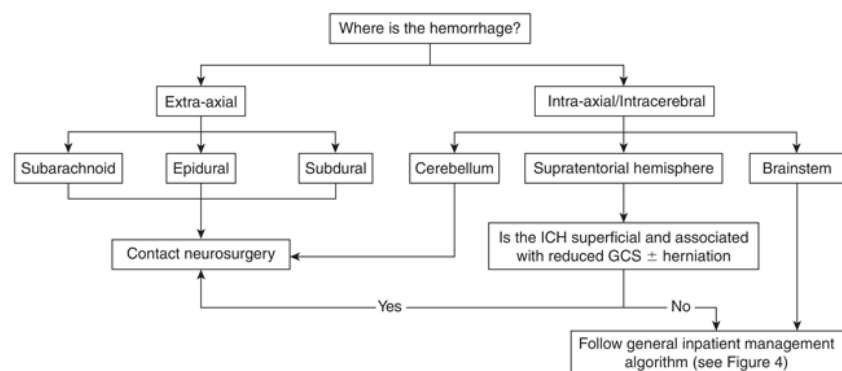
Abbreviations: BA = basilar artery; CBC = complete blood count; CT = computed tomography; EVT = endovascular thrombectomy; ICA = intracranial artery; INR = International Normalized Ratio; IV = intravenous; MCA = middle cerebral artery

Figure 2: Management of the Patient with a Nonischemic Lesion on CT Scan



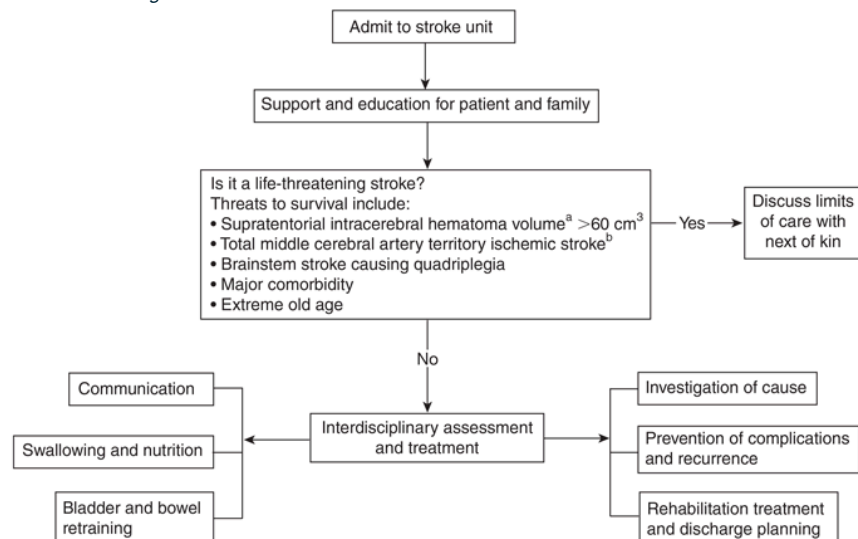
Abbreviations: CT = computed tomography

Figure 3: Approach to a Finding of Intracranial Hemorrhage on CT Scan



Abbreviations: GCS = Glasgow Coma Scale; ICH = intracerebral hemorrhage

Figure 4: General Management of the Stroke Patient



^[a] Hematoma volume (cm³) = (A×B×C)/2 where A = largest diameter (cm) on CT scan, B = diameter (cm) perpendicular to A, C = number of 1-cm scan slices showing hematoma.^{[14][15]}

^[b] Hemiparesis and hemianopia with either aphasia or visuospatial deficit.^[17]

Drug Tables

Table 5: Drugs Used to Treat Acute Ischemic Stroke

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
Drug Class: Antiplatelet Agents			
ASA Aspirin, generics <\$10	Initial: 160 mg PO Maintenance: 80–325 mg daily PO	Nausea, vomiting, hemorrhage, hypersensitivity reactions (rarely bronchospasm).	If alteplase is to be administered, wait 24 h before initiating ASA. If alteplase is not used, start ASA as soon as ICH is excluded by CT scan. Use enteric-coated formulations for patients who can swallow. Give via nasogastric tube or as a suppository to dysphagic patients.
clopidogrel Plavix, Clopidogrel, other generics \$10–25	Initial: 300–600 mg PO Maintenance: 75 mg daily PO	Hemorrhage, skin rash, diarrhea.	
ticagrelor Brilinta, generics \$10–25	Initial: 180 mg PO Maintenance: 90 mg BID PO	Hemorrhage, rash, dizziness, diarrhea, shortness of breath.	Consider in patients who have CYP2C19 loss-of-function allele.
Drug Class: Thrombolytics			
alteplase Activase rt-PA \$1,600/50 mg vial	0.9 mg/kg (maximum 90 mg) IV over 60 min Give 10% of the total dose as an initial bolus over 1 min	Superficial bleeding, internal bleeding (e.g., GI, genitourinary, respiratory tract, retroperitoneal), ICH . Cerebral edema, cerebral herniation, seizures and new onset ischemic	Review inclusion and exclusion criteria before administering (see Table 3). No anticoagulant or antiplatelet drugs for 24 h after administration.

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
		strokes may occur and be life-threatening.	
		Hypersensitivity reactions, including orolingual angioedema.	
tenecteplase TNKase \$3,350/(50 mg vial)	0.25mg/kg (maximum 25mg) IV over 5 seconds ^[28]	Superficial bleeding, internal bleeding (e.g., GI, genitourinary, respiratory tract, retroperitoneal), ICH . Cerebral edema, cerebral herniation, seizures and new onset ischemic strokes may occur and be life-threatening.	Review inclusion and exclusion criteria before administering (see Table 3). No anticoagulant or antiplatelet drugs for 24 h after administration.
		Hypersensitivity reactions, including orolingual angioedema.	

^[a] Cost of a 30-day supply unless otherwise specified; includes drug cost only.
Abbreviations: ASA = acetylsalicylic acid; CT = computed tomography; GI = gastrointestinal; ICH = intracranial hemorrhage; IV = intravenous

Table 6: Drugs Used to Treat Subarachnoid Hemorrhage

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
Drug Class: Calcium Channel Blockers			
nimodipine Nimotop \$2,730	60 mg Q4H PO × 21 days 30 mg Q2H PO may be used if hypotension significant	Hypotension (5%), nausea, bradycardia, rash.	Crushing tablets for nasogastric administration is not recommended since bioavailability reduces with time. If necessary, crush and administer immediately.

^[a] Cost of 21-day supply; includes drug cost only.

Suggested Readings

[Embersen J, Lees KR, Lyden P et al. Effect of treatment delay, age, and stroke severity on the effects of intravenous thrombolysis with alteplase for ischemic stroke: a meta-analysis of individual patient data from randomized trials. *Lancet* 2014;384\(9958\):1929-35.](#)

[Goyal M, Menon BK, van Zwam WH et al. Endovascular thrombectomy after large-vessel ischaemic stroke: a meta-analysis of individual patient data from five randomised trials. *Lancet* 2016;387\(10029\):1723-31.](#)

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